

Python Viva Questions

What is Python?

- Python is a high-level, interpreted programming language known for its readability and simplicity. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming.

What are the key features of Python?

- Easy-to-read syntax
- Dynamic typing
- Interpreted nature
- Extensive standard libraries
- Support for multiple programming paradigms
- Cross-platform compatibility

What is the difference between list and tuple in Python?

- list is mutable, meaning its contents can be changed after creation.
tuple is immutable, meaning once it is created, it cannot be modified.
Tuples can be used as keys in dictionaries, while lists cannot.

Data Structures

How do you create a dictionary in Python?

- You can create a dictionary using curly braces {}, with key-value pairs separated by colons:
`my_dict = {'key1': 'value1', 'key2': 'value2'}`

What is a set in Python, and how is it different from a list?

- A set is an unordered collection of unique elements. Unlike lists, sets do not allow duplicate values and do not maintain any order of elements.

How do you handle missing values in a pandas DataFrame?

- Missing values can be handled using methods like `dropna()` to remove them or `fillna()` to replace them with a specific value or method.

Functions and Modules

What are lambda functions in Python?

- Lambda functions are small anonymous functions defined using the `lambda` keyword. They are often used for short-term operations that are not meant to be reused.

`square = lambda x: x * x`

What is the purpose of the `__init__` method in Python classes?

- The `__init__` method is a special method that is called when an instance of the class is created. It is used to initialize the object's attributes.

How do you import a module in Python?

- You can import a module using the `import` keyword. For example:
`import math`

Object-Oriented Programming

What is inheritance in Python?

- Inheritance is a mechanism where a new class (child class) inherits attributes and methods from an existing class (parent class). It allows for code reuse and the creation of a hierarchical class structure.

What are class methods and static methods?

- Class methods are methods that are bound to the class and not the instance. They are defined using the `@classmethod` decorator and take `cls` as the first parameter.

- Static methods do not take a special first parameter and are defined using the `@staticmethod` decorator. They are used for utility functions that don't need access to class or instance-specific data.

What is OOP and how is it implemented in Python?

- Object-oriented programming (OOP) is a programming paradigm based on objects and classes. Python implements OOP with features like inheritance, encapsulation, and polymorphism.

What is the self parameter in Python classes?

- The self parameter refers to the instance of the class itself, allowing access to attributes and methods of the class.

What is inheritance? Give an example.

- Inheritance allows a class (child class) to inherit attributes and methods from another class (parent class).

```
class Animal:
```

```
    def speak(self):  
        return "Animal speaks"
```

```
class Dog(Animal):
```

```
    def bark(self):  
        return "Dog barks"
```

Error Handling

How do you handle exceptions in Python?

- Exceptions are handled using try, except, else, and finally blocks. For example:

```
try:  
    result = 10 / 0
```

```
except ZeroDivisionError:
    print("Cannot divide by zero.")
finally:
    print("This block always executes.")
```

What is the difference between raise and assert in Python?

- raise is used to explicitly throw an exception, while assert is used for debugging purposes to test if a condition is true. If the condition is false, assert will raise an AssertionError.

What are decorators in Python?

- Decorators are functions that modify or extend the behavior of other functions or methods. They are applied using the @decorator_name syntax.

What is a generator in Python?

- A generator is a special type of iterator that is defined using a function with yield statements. Generators are used to create iterators in a more memory-efficient way.

How does Python manage memory?

- Python uses automatic memory management with a built-in garbage collector that handles memory allocation and deallocation. The garbage collector uses reference counting and cyclic garbage collection to manage memory.

What are Python's built-in data types?

- Python's built-in data types include:
 - Numeric: int, float, complex
 - Sequence: list, tuple, range
 - Text: str
 - Mapping: dict

- Set: set, frozenset
- Boolean: bool
- Binary: bytes, bytearray, memoryview

What are lists and tuples in Python?

- Lists are mutable (modifiable) sequences, whereas tuples are immutable (non-modifiable). This means you can change a list after its creation but cannot change a tuple.

What is a dictionary in Python?

- A dictionary is an unordered collection of key-value pairs. It is mutable and indexed by keys, allowing for efficient data retrieval.

Explain the difference between == and is in Python.

- == checks for value equality (whether two objects have the same value), while is checks for identity (whether two references point to the same object in memory).

What are Python decorators?

- Decorators are functions that modify the behavior of another function. They are often used to add functionality, such as logging or access control, to existing functions.

What is a lambda function?

- A lambda function is an anonymous function defined with the lambda keyword. It can take any number of arguments but can only have one expression. It's often used for short, throwaway functions.

What is the purpose of the self parameter in class methods?

- The self parameter refers to the instance of the class itself. It is used to access variables and methods associated with the instance.

What is a generator in Python?

- A generator is a special type of iterator that is defined using a function and the yield statement. Generators allow for lazy evaluation, producing items one at a time and only when requested.

Explain the concept of list comprehensions.

- List comprehensions provide a concise way to create lists. They consist of an expression followed by a for clause and can include optional if conditions.

What are Python's built-in exceptions?

- Python has several built-in exceptions such as ValueError, TypeError, IndexError, KeyError, ZeroDivisionError, etc. These exceptions help in error handling and debugging.

How does Python handle memory management?

- Python uses a private heap space to manage memory, and it employs a built-in garbage collector to reclaim memory from objects that are no longer in use.

What are modules and packages in Python?

- A module is a file containing Python code that can define functions, classes, and variables. A package is a collection of modules organized in a directory hierarchy, allowing for better organization of related modules.

What is the difference between deepcopy and copy?

- The copy module provides shallow and deep copy methods. A shallow copy creates a new object but inserts references into it to the objects found in the original. A deep copy creates a new object and recursively adds copies of nested objects.

How do you handle exceptions in Python?

- Exceptions are handled using the try, except, finally, and else blocks. Code that may raise an exception is placed in the try block, while the handling code is placed in the except block.

What are the benefits of using Python?

- Benefits include ease of learning, extensive libraries, community support, versatility, and cross-platform compatibility.

What are the data types supported by Python?

- Common data types include:
 - Numeric: int, float, complex
 - Sequence: list, tuple, range
 - Text: str
 - Mapping: dict
 - Set: set, frozenset
 - Boolean: bool

What is the difference between lists and tuples?

- Lists are mutable (can be modified), while tuples are immutable (cannot be modified).

Explain how Python handles memory management.

- Python uses a private heap space and a garbage collector to manage memory automatically.

Control Structures

What are conditional statements in Python?

- Conditional statements (if, elif, else) allow you to execute different blocks of code based on certain conditions.

Explain loops in Python.

- Python supports for and while loops. for loops iterate over a sequence, while while loops continue until a condition is false.

What is the purpose of the break and continue statements?

- break exits the loop, while continue skips the current iteration and moves to the next one.

What are functions in Python?

- Functions are reusable blocks of code that perform a specific task. They can take inputs (arguments) and return outputs.

How do you define a function in Python?

- You define a function using the def keyword followed by the function name and parentheses. For example:

```
def my_function(param):  
    return param * 2
```

What is the difference between *args and **kwargs?

- *args is used to pass a variable number of non-keyword arguments to a function, while **kwargs is used to pass a variable number of keyword arguments.

What is a module in Python?

- A module is a file containing Python code (functions, classes, variables) that can be imported and used in other Python programs.

What are decorators in Python?

- Decorators are functions that modify the behavior of other functions or methods. They are often used for logging, authentication, and access control.

Explain the concept of generators.

- Generators are a type of iterable that generate values on the fly using the yield keyword, allowing for efficient memory usage.

What is exception handling in Python?

- Exception handling is a mechanism to handle errors gracefully using try, except, finally, and else blocks.

How do you read and write files in Python?

- You can use the open() function to read and write files. Example for writing:

with open('file.txt', 'w') as f:

f.write("Hello, World!")

What are list comprehensions? Provide an example.

- List comprehensions provide a concise way to create lists. Example:
squares = [x**2 for x in range(10)]

What are Python's built-in functions? Give some examples.

- Built-in functions include print(), len(), type(), str(), int(), list(), etc.

What is the purpose of the with statement?

- The with statement simplifies exception handling by encapsulating common preparation and cleanup tasks in so-called context managers.

How do you install external libraries in Python?

- External libraries can be installed using pip, the Python package manager, with the command pip install package_name.

Explain the difference between shallow copy and deep copy.

- A shallow copy creates a new object but inserts references to the original objects, while a deep copy creates a new object and recursively copies all objects.

What is the Global Interpreter Lock (GIL) in Python?

- The GIL is a mutex that protects access to Python objects, preventing multiple threads from executing Python bytecode simultaneously.